Linux File Permissions and Access Control Lab

1. What I Wanted to Learn

I wanted to learn how Linux file permissions work and how they can be used to control who has access to information. To practice, I set up a small lab in Ubuntu and tested access using a separate user account.

2. My Lab Setup

I ran Ubuntu Linux in VMware Fusion on my MacBook. Inside Ubuntu, I created a folder called cybersecurity-lab with three subfolders:

HR: Employee records

Finance: Budget information

Public: Company welcome information

I also created a separate Linux user named employee1 so I could test the permissions without using my main account.

3. Setting File Permissions

I used the Linux chmod command to set permissions for the three files.

File

Permission

What it does

HR employee records

600

Allows the owner to read and write the file.

Finance budget

600

Allows the owner to read and write the file.

Public welcome file

644

Allows the owner to read and write the file while other users can read it.

I also learned that a user needs permission to pass through the directories leading to a file. File permissions alone don't determine whether someone can reach it.

4. Testing Access

After setting up the files, I used the employee1 account to test access.

HR employee records: Access was denied. The test account could not read the file.

Finance budget: Access was denied. The test account could not read the file.

Public welcome file: Access was initially denied, even though the file was set to allow other users to read it. I investigated the issue and adjusted the home directory's traversal permissions. After that change, the test account could read the Public file successfully.

5. What I Learned

This lab helped me understand how Linux permissions work in practice. I learned how to create a separate user, change file permissions, test access, and troubleshoot a permissions problem.

One thing that stood out was that access depends on more than the file itself. Directory permissions can also prevent a user from reaching a file, even when that file allows them to read it.

I also learned why it's important to be careful when changing permissions on a home directory. In a real environment, I'd look at using a dedicated shared directory or group permissions instead of allowing unnecessary access to a personal home directory.

6. Results

The access tests produced the expected results for the HR and Finance files: employee1 was denied access to both. The Public file became readable after I corrected the directory traversal permissions.

This gave me practical experience with Linux file permissions, user access, and basic troubleshooting.

7. What I'd Like to Practice Next

For my next lab, I'd like to practice managing access with Linux groups, reviewing permissions across an entire directory, and setting up a shared folder without exposing a user's personal home directory.

These are useful skills to keep building as I work toward cybersecurity and Identity and Access Management (IAM) roles.

Tools Used

Ubuntu Linux

VMware Fusion

Linux Terminal